

A certified bracket for the Erdos-Herzog-Piranian infimum

[OPERATOR] author line and affiliation. [OPERATOR] venue. [OPERATOR] AI-involvement disclosure — the norm on this problem's forum is currently contested and an explicit statement is expected.

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Abstract

For a nonconstant monic polynomial f with all roots in $[-1,1]$, Erdos, Herzog and Piranian (1958) asked for the extremes of the measure of the set where f has modulus below 1. Writing μ for the root measure and $U_\mu(x)$ for the logarithmic potential, the integral of $\log(1/|x-t|)$ against μ , the quantity is the measure of the set where $U_\mu < 0$. We prove, in outward-rounded interval arithmetic with a machine-checkable certificate,

$$1.828 \leq \inf_{\mu} |\{U_\mu < 0\}| \leq 1.8344304971959906.$$

The lower bound uses no tail estimate, assumes no minimizer, and argues by no contradiction: it is a direct forcing argument whose only analytic input beyond interval arithmetic is the symmetry of the mutual energy, that the integral of U_ν against μ equals the integral of U_μ against ν . We further answer affirmatively, for every ϵ in $(0, 0.1]$, the model problem posed as Problem 4.1 in Tao's notes on this problem, and we certify three dual measures posted by the community which had previously been validated by sampling only.

Three AI-assisted proofs claiming the exact value $D = 1.83443047576266171109\dots$ are outstanding at the time of writing (Section 5). THIS PAPER ASSUMES NONE OF THEM and adjudicates none of them; the bracket above is unconditional.

1. Setting

Every nonconstant monic f of degree d with roots in $[-1,1]$ corresponds to the root measure μ , the average of point masses at the roots, and $f(x)$ has modulus below 1 exactly where $U_\mu(x) > 0$ after the standard normalization; passing to general probability measures on $[-1,1]$ is the usual closure. The two extremes are independent problems. The supremum is the subject of a separate certified program of ours and is not treated here.

2. The upper bound

THEOREM 1. For the probability measure $\mu = A \delta_{-1} + \int_0^1 f(y) dy$ on $[a,1]$, where δ_{-1} is the unit mass at -1 , with the exact rational parameters

$$a = 0.804462, A = 0.8245218, \\ f(y) = [1 - A \sqrt{2(1+a)} / (1+y)] / (\pi \sqrt{(y-a)(1-y)}),$$

the set where $U_\mu < 0$ is the interval (x_L, x_R) with the point -1 removed,

$$\begin{aligned} x_L &\text{ in } [-1.8081074413361424, -1.808107441336006], \\ x_R &\text{ in } [0.02632305585964334, 0.026323055859847976], \end{aligned}$$

so the infimum is at most 1.8344304971959906.

The potential off the support is $c + g(x) - A g_\Omega(x, -1)$ in the Joukowski coordinate, with c the constant on-support level and g the Green's functions of the complement; the certificate verifies $c \geq 0$ (certified c in $[9.315e-8, 9.315e-8]$), the nonnegativity of the density, and the two root enclosures. The set structure follows from three one-line lemmas: the sign of U' left of -1 , concavity on $(-1, a)$, and positivity right of 1 . Classical inputs: the arcsine potential identity and the balayage of a point mass onto an interval (Saff-Totik), each bridged numerically in the lab.

3. The lower bound

THEOREM 2. The infimum is at least 1.828.

Under the standard reduction — μ normalized with support inside $\{-1\}$ union $[0, 1]$, and the interval $(-\sqrt{2}, 0)$ contained in E , the set where $U_\mu > 0$ — let a_0 be the infimum of the component of E containing $(-\sqrt{2}, 0)$. Then $U_\mu(a_0) \leq 0$ by lower semicontinuity and a_0 lies in $[-2, -\sqrt{2}]$.

THE SELECTOR. If ν is a finite atomic measure, a unit mass at a_0 plus nonnegative weights w_i at points p_i , and $U_\nu > 0$ on the support of μ , then the symmetry of the mutual energy forces at least one p_i to lie in E .

THE COMB. If $a_0 \leq -\text{cap}$ the component already has length at least cap . Otherwise, with $R = \text{cap} - |a_0|$, the certificate exhibits for every b in $[0, R]$ a measure ν_b — the mass at a_0 , a heavy atom at b , and descending teeth at $s_j - b$ — with $U_{\{\nu_b\}} > 0$ on $\{-1\}$ union $[0, 1]$, hence on the support of μ . The atoms travel at unit speed along pairwise disjoint images, all disjoint from $(a_0, 0)$, so the selector gives $|E| \geq |a_0| + R = \text{cap}$.

WHAT IS CERTIFIED. The parameter a_0 is tiled by 86 boxes over the full range $[-\text{cap}, -\sqrt{2}]$; each carries one comb, and 1,350 b-boxes in total, each with one frozen weight vector obtained from a floating-point linear program and then **VERIFIED** by intervals, never trusted. Two monotonicity facts collapse the two-parameter problem to thin one-dimensional convex problems: $U_\nu(x)$ increases in a_0 for $x > a_0$, and in the shifted coordinate $y = x + b$ the teeth do not move while the remainder increases in b . Convex pole sums are bounded below by interval tangent lines between consecutive poles and bisected to positivity. Worst certified margin over all boxes: $4.116e-7$; runtime 265 seconds.

An independent checker (verify-forcing.js, sharing no code with the certifier) re-derives the tiling and weight structure from the certificate file and searches for counterexamples in floating point; it reports no violation.

REMARK (the ceiling of the method). The same forcing at the conjectured optimum can move the free atom only as far as x_R , about 0.0263, so this route cannot exceed the Theorem 1 value —

the method is exact in principle. The comb family in particular is exhausted near 1.828: its certified margin has thinned to $4.12\text{e-}7$ and finer boxes cannot buy the next thousandth. Passing it requires demanding positivity only on the minimizer's own support, which is a different construction rather than a longer run.

4. Tao's Problem 4.1, and a mechanism

THEOREM 3. For every ϵ in $(0, 0.1]$ the explicit measure λ^ϵ is positive with $U_\lambda \geq 0$ on all of $[-1, 1]$; the two-interval scenario is therefore excluded for the whole range.

The certificate covers $[1\text{e-}12, 0.1]$ with 624,275 interval chunks, each carrying its own rational parameter, and a separate sliver lemma closes $(0, 1\text{e-}12]$ by the substitution $t = \zeta_1 - 1$, under which the $\epsilon \rightarrow 0$ singular pole cancels algebraically rather than numerically.

THE DELTA-MECHANISM. As ϵ tends to 0 the margin at $x = -1$ tends to a constant δ determined by the level defect of the primal constants. With rounded constants δ may have either sign, and if it is negative the family PROVABLY FAILS for ϵ below roughly $|\delta|/0.21$. This was not anticipated: the certifier returned a decisively negative window and the mechanism was identified afterwards to explain it. Constructions built from midpoint decimals will meet this wall; all constructions posted on the problem's forum sit far above it. Our test suite keeps the failure as a control — putting the defect on the wrong side must make the certificate refuse.

THEOREM 4 (community measures). The three dual measures posted on the forum, previously validated by sampling, are positive with $U_\lambda \geq 0$ on $[-1, 1]$; certified off-atom minima $1.243\text{e-}5$, $2.653\text{e-}6$, $3.959\text{e-}6$. Positivity is established by adaptive tangent envelopes on each gap between support points — no quadrature and no grid, which is what sampling could not supply.

5. Relation to the outstanding claims

At the time of writing the problem's forum lists three claimed full proofs, all AI-assisted, each reporting the same value $D = 1.83443047576266171109\dots$: Darvas, Peng and Runzhou Tao (2026-07-15, GPT-5.5 Pro in a prover-verifier framework); Shouqiao Wang (GPT-5.6 Sol); and Cristian Budala (2026-08-24, GPT-5.6 Sol). The forum states that appearing there "is no guarantee of proof correctness, and does not mean that anyone associated with this site has examined any part of the proof."

We have separately re-verified the COMPUTATIONAL APPENDIX of the Darvas-Peng-Tao manuscript — existence and local uniqueness of the extremal triple by an interval Krawczyk operator, and all thirty printed decimals of D — and we did NOT audit its analytic core. Nothing in the present paper depends on any of the three. Our bracket contains D , which is corroboration and not confirmation.

[OPERATOR] decision. Whether to submit this while three claimed solutions are outstanding, and if so how to frame the contribution — the unconditional bracket, the tail-free forcing method,

and Problem 4.1 — is a judgment call.

6. Reproduction

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node tools/run-lemniscate.js  
node instruments/lemniscate/forcing.js 1.828 certs/erdos1038-forcing-1.828.json  
node instruments/lemniscate/battery.js
```

The first recomputes Theorems 1, 3 and 4 and re-verifies the forcing record; the second rebuilds that record from scratch; the third runs the checks and the red controls, every one of which must fire. The certifier is `instruments/interval` — outward-rounded interval arithmetic with `BigInt` rational falsifiers — the same one behind this repository's other certified theorems. Records: `certs/erdos1038-inf.json` and `certs/erdos1038-forcing-1.828.json`.